

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce Department of Information Technology (B.Sc.I.T Semester V)

Applied Business Analytics

M2: Practical 9

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Aim:- Hypothesis Testing:

A. Formulate null and alternative hypotheses related to a business problem, conduct a hypothesis test using appropriate statistical tests, and interpret the Results.

1. Load the supermarket_sales.csv dataset.:

```
import pandas as pd
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
df = pd.read_csv("supermarket_sales.csv")
```

2. Display the first five records.

```
print("First five records:")
print(df.head())
```

Output:

First five records:									
	Invoice ID	Branch	City	Customer type	Gender				
0	750-67-8428	A	Yangon	Member	Female				
1	226-31-3081	C	Naypyitaw	Normal	Female				
2	631-41-3108	A	Yangon	Normal	Male				
3	123-19-1176	A	Yangon	Member	Male				
4	373-73-7910	A	Yangon	Normal	Male				

	Product line	Unit price	Quantity	Tax 5%	Total	Date
0	Health and beauty	74.69	7	26.1415	548.9715	1/5/2019
1	Electronic accessories	15.28	5	3.8200	80.2200	3/8/2019
2	Home and lifestyle	46.33	7	16.2155	340.5255	3/3/2019
3	Health and beauty	58.22	8	23.2880	489.0480	1/27/2019
4	Sports and travel	86.31	7	30.2085	634.3785	2/8/2019

	Time	Payment	cogs	gross margin percentage	gross income	Rating
0	13:08	Ewallet	522.83	4.761905	26.1415	9.1
1	10:29	Cash	76.40	4.761905	3.8200	9.6
2	13:23	Credit card	324.31	4.761905	16.2155	7.4
3	20:33	Ewallet	465.76	4.761905	23.2880	8.4
4	10:37	Ewallet	604.17	4.761905	30.2085	5.3

3. Select the revenue column.

```
revenue = df["Total"]
print("\nRevenue:")
print(revenue)
```

Output:

```
Revenue:
0      548.9715
1      80.2200
2     340.5255
3     489.0480
4     634.3785
...
995    42.3675
996   1022.4900
997    33.4320
998    69.1110
999    649.2990
Name: Total, Length: 1000, dtype: float64
```

4. Check for missing values.

```
print("\nMissing values in revenue:")
print(revenue.isnull().sum())
# Remove missing values if present
revenue = revenue.dropna()
```

Output:

```
Missing values in revenue:
0
```

5. Calculate the mean revenue.

```
mean_revenue = revenue.mean()
print("\nMean Revenue:", mean_revenue)
```

Output:

```
Mean Revenue: 322.966749
```

6. Assume that the expected average revenue is ₹1,500
expected_mean = 1500

7. Write the null hypothesis:

H0: $\mu = 1500$

8. Write the alternative hypothesis:

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H1: $\mu \neq 1500$

Output:

```
# 7. Alternative Hypothesis
# H1: pA != pB

# 8. Significance level
alpha = 0.05

✓ 0.0s
```

9. Set the significance level as 5%.

alpha = 0.05

10. Perform a one-sample t-test.

```
t_stat, p_value = stats.ttest_1samp(
    revenue,
    expected_mean
)
```

11. Calculate the p-value.

```
print("\nT-statistic:", t_stat)
print("P-value:", p_value)
```

Output:

```
T-statistic: -151.37567896776733
P-value: 0.0
```

```
import pandas as pd
from statsmodels.stats.proportion import proportions_ztest
import matplotlib.pyplot as plt
data = pd.read_csv("ab_testing.csv")
# Convert Conversion column into numeric values
# If values are 0 and 1 stored as text, this will convert them
data["Conversion"] = pd.to_numeric(data["Conversion"],
    errors="coerce")
```

2. Display the data

```
print("A/B Testing Dataset:")
print(data)
```

Output:

A/B Testing Dataset:			
	User	Group	Conversion
0	14292	B	NaN
1	11682	A	NaN
2	19825	A	NaN
3	16080	B	NaN
4	18851	A	NaN
...	...	...	...
4995	16360	B	NaN
4996	18084	B	NaN
4997	12063	A	NaN
4998	18647	B	NaN
4999	16686	A	NaN

```
# Check for missing values
data = data.dropna(subset=["Conversion"])
```

3. Calculate conversion rate for Group A

```
group_A = data[data["Group"] == "A"]
users_A = len(group_A)
conversions_A = group_A["Conversion"].sum()
```

12. Compare the p-value with 0.05.

13. State whether the null hypothesis is rejected or not rejected.

```
if p_value < alpha:
    print("Reject the Null Hypothesis")
else:
    print("Do not Reject the Null Hypothesis")
```

Output:

```
Reject the Null Hypothesis
```

14. Write the conclusion in simple business terms.

Conclusion:

The test shows that the average revenue is significantly different from ₹1,500. Therefore, the business should not assume that its average revenue is ₹1,500 based on the given data.

B. Perform A/B testing on a website or marketing campaign to evaluate the effectiveness of different strategies and make data-driven decisions

1. Load the dataset

```
rate_A = conversions_A / users_A
print("\nGroup A Conversion Rate:", rate_A)
print("Group A Conversion Rate:", rate_A * 100, "%")
```

Output:

```
Group A Conversion Rate: nan
Group A Conversion Rate: nan %
C:\Users\itlab\AppData\Local\Temp\ipykernel_
rate_A = conversions_A / users_A
```

4. Calculate conversion rate for Group B

```
group_B = data[data["Group"] == "B"]
users_B = len(group_B)
conversions_B = group_B["Conversion"].sum()
rate_B = conversions_B / users_B
print("\nGroup B Conversion Rate:", rate_B)
print("Group B Conversion Rate:", rate_B * 100, "%")
```

Output:

```
Group B Conversion Rate: nan
Group B Conversion Rate: nan %
C:\Users\itlab\AppData\Local\Temp\ipykernel_
rate_B = conversions_B / users_B
```

5. Compare conversion rates

```
print("\nComparison:")
print("Group A:", rate_A * 100, "%")
print("Group B:", rate_B * 100, "%")
if rate_A > rate_B:
    print("Group A has a higher conversion rate.")
elif rate_B > rate_A:
    print("Group B has a higher conversion rate.")
else:
```

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```
print("Both groups have the same conversion rate.")
```

Output:

```
Comparison:
Group A: nan %
Group B: nan %
Both groups have the same conversion rate.
```

6. Null Hypothesis

```
print("\nH0: pA = pB")
```

Output:

```
H0: pA = pB
```

7. Alternative Hypothesis

```
print("H1: pA != pB")
```

Output:

```
H1: pA != pB
```

8. Significance level

```
alpha = 0.05
```

```
print("Significance Level:", alpha)
```

Output:

```
Significance Level: 0.05
```

9. Perform two-proportion test

```
conversions = [conversions_A, conversions_B]
```

```
users = [users_A, users_B]
```

```
z_stat, p_value = proportions_ztest(conversions, users)
```

Output:

```
c:\Users\itlab\AppData\Local\Programs\Python\Py
prop = count * 1. / nobs
c:\Users\itlab\AppData\Local\Programs\Python\Py
p_pooled = np.sum(count) * 1. / np.sum(nobs)
c:\Users\itlab\AppData\Local\Programs\Python\Py
nobs_fact = np.sum(1. / nobs)
```

10. Calculate and display p-value

```
print("\nZ-statistic:", z_stat)
```

```
print("P-value:", p_value)
```

Output:

```
Z-statistic: nan
P-value: nan
```

11. Compare p-value with 0.05

```
print("\nComparison of p-value with 0.05:")
```

```
if p_value < alpha:
```

```
    print("p-value < 0.05")
```

```
else:
```

```
    print("p-value >= 0.05")
```

Output:

```
Comparison of p-value with 0.05:
p-value >= 0.05
```

12. Decision

```
if p_value < alpha:
```

```
    print("Decision: Reject the Null Hypothesis (H0)")
```

```
else:
```

```
    print("Decision: Do not reject the Null Hypothesis (H0)")
```

Output:

```
Decision: Do not reject the Null Hypothesis (H0)
```

13. Create bar chart

```
groups = ["Group A", "Group B"]
```

```
conversion_rates = [rate_A * 100, rate_B * 100]
```

```
plt.bar(groups, conversion_rates)
```

```
plt.xlabel("Group")
```

```
plt.ylabel("Conversion Rate (%)")
```

```
plt.title("A/B Testing - Conversion Rate Comparison")
```

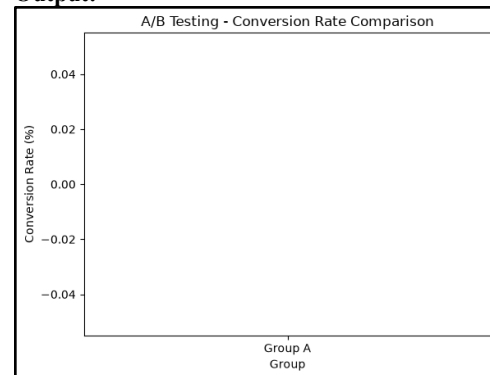
```
# Display values on bars
```

```
for i, rate in enumerate(conversion_rates):
```

```
    plt.text(i, rate, f'{rate:.2f}%', ha="center", va="bottom")
```

```
plt.show()
```

Output:



14. Write the conclusion based on the test result.

Conclusion:

The conversion rate of Group A and Group B was calculated and compared using a two-proportion z-test at a 5% significance level. Based on the p-value, the null hypothesis is not rejected if the p-value is greater than 0.05. Therefore, there is not enough statistical evidence to conclude that the conversion rates of Group A and Group B are significantly different.